

ECEN 743: Reinforcement Learning

RL with Function Approximation

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References

- [SB, Chapter 9-11]
- [BDP, Chapter 6]
- [MMT, Chapter 12]

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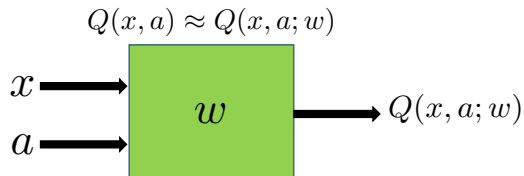
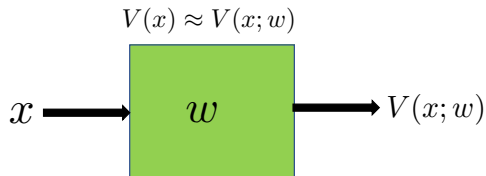
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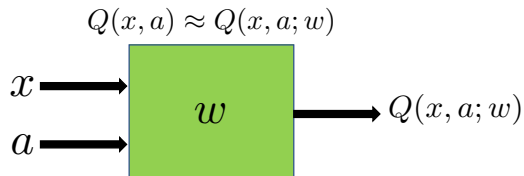
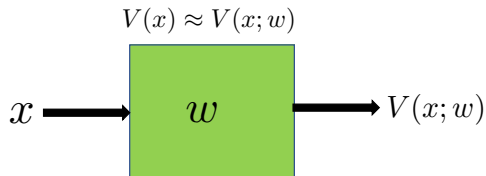
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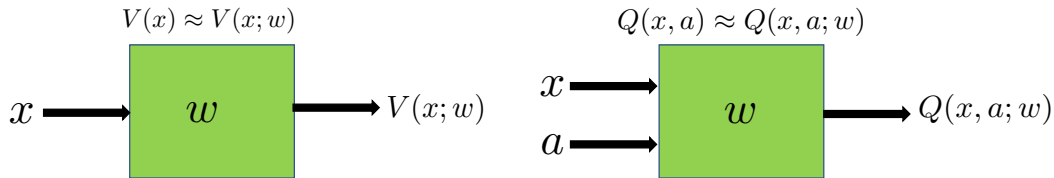
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Function approximation will extrapolate/generalize the value/policy from seen states to unseen states

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- **Example:** Fourier (sine/cosine) series approximation

$$f(x) \approx \sum_{i=1}^d w_i \cos(i\nu_0 x)$$

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$$\phi(s) = (s_1, s_2)^\top, \quad \phi(s) = (1, s_1, s_2, s_1 s_2)^\top$$

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- ▶ Assume that $\mathcal{S} \subset \mathbb{R}^n$, and $\phi(s) \in \mathbb{R}^d$. Radial basis functions are defined as

$$\phi_i(s) = \exp\left(-\frac{\|s - \mu_i\|^2}{2\sigma_i^2}\right),$$

where μ_i, σ_i are fixed a priori

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For each $s \in \mathcal{S}$, $\phi(s) \in \mathbb{R}^d$, $\phi(s) = (\phi_1(s), \phi_2(s), \dots, \phi_d(s))^T$

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- $(\phi_i)_{i=1}^d$ are often assumed to be independent. So, they are also called basis functions

Matrix Notation

- We can write $V(w)(s) = (\Phi w)(s)$, where

$$\Phi = \begin{bmatrix} \phi_1(s_1) & \phi_2(s_1) & \cdots & \phi_d(s_1) \\ \phi_1(s_2) & \phi_2(s_2) & \cdots & \phi_d(s_2) \\ \vdots & & \ddots & \vdots \\ \phi_1(s_{|S|}) & \phi_2(s_{|S|}) & \cdots & \phi_d(s_{|S|}) \end{bmatrix} = \begin{bmatrix} | & | & \cdots & | \\ \phi_1 & \phi_2 & \cdots & \phi_d \\ | & | & \cdots & | \end{bmatrix}$$

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- The actual value functions may be in an $|S|$ -dimensional space $\mathcal{V} \subset \mathbb{R}^{|S|}$

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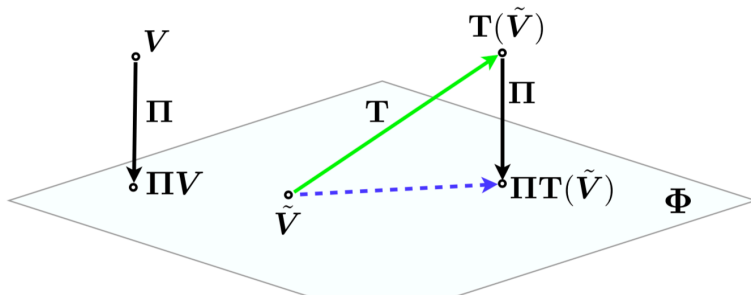
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- Pythagorean theorem: If V_1 and V_2 are orthogonal, $\|V_1 + V_2\|^2 = \|V_1\|^2 + \|V_2\|^2$

Approximate Dynamic Programming

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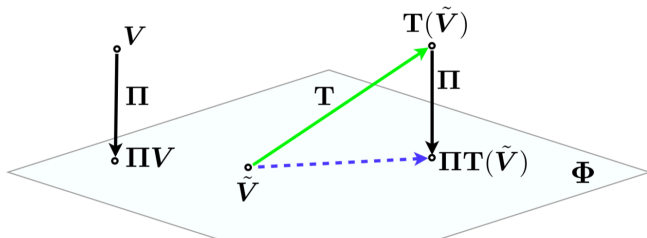
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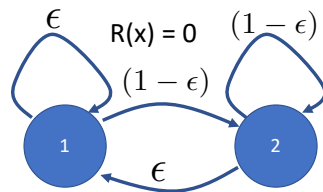
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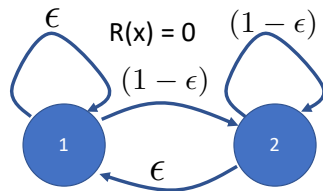
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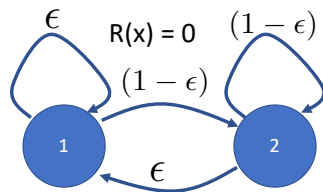
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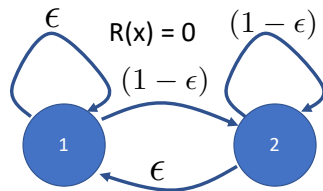


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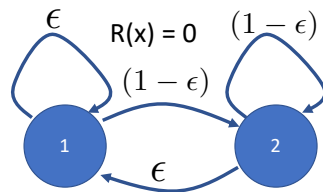
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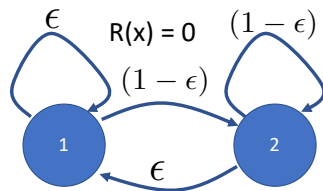


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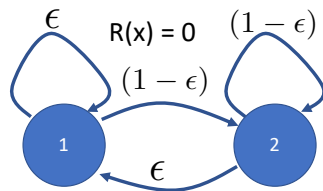
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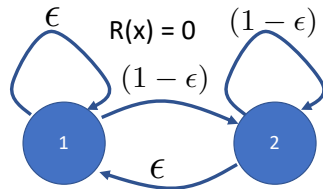
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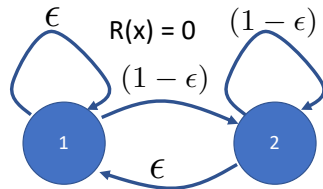


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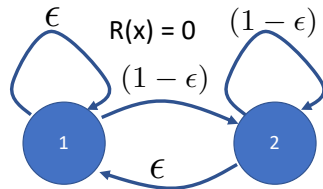
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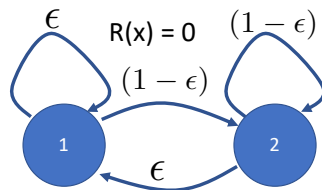
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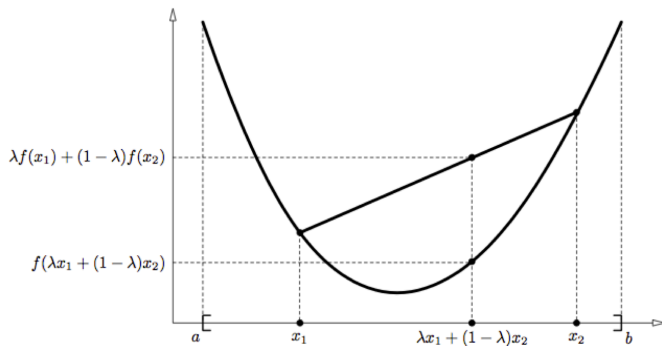
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Proof: Observe that $\bar{V}_\pi - \Pi V_\pi$ and $V_\pi - \Pi V_\pi$ are orthogonal vectors w.r.t. the inner product $\langle V_1, V_2 \rangle = \sum_s \mu(s) V_1(s) V_2(s)$. Now,

$$\begin{aligned} \|\bar{V}_\pi - V_\pi\|_{2,\mu}^2 &= \|\bar{V}_\pi - \Pi V_\pi + \Pi V_\pi - V_\pi\|_{2,\mu}^2 \\ &= \|\bar{V}_\pi - \Pi V_\pi\|_{2,\mu}^2 + 2 \langle \bar{V}_\pi - \Pi V_\pi, \Pi V_\pi - V_\pi \rangle + \|\Pi V_\pi - V_\pi\|_{2,\mu}^2 \\ &= \|\Pi T_\pi \bar{V}_\pi - \Pi T_\pi V_\pi\|_{2,\mu}^2 + \|\Pi V_\pi - V_\pi\|_{2,\mu}^2 \\ &\leq \gamma^2 \|\bar{V}_\pi - V_\pi\|_{2,\mu}^2 + \|\Pi V_\pi - V_\pi\|_{2,\mu}^2 \end{aligned}$$

Approximation Error

- ΠT_π is a contraction mapping. So, it has a unique fixed point $\bar{V}_\pi$. The iteration $V_{k+1} = \Pi T_\pi V_k$ will converge to $\bar{V}_\pi$
- What is the approximation error $\|V_\pi - \bar{V}_\pi\|_{2,\mu}$?

Proposition (Approximation Error)

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This implies, $\|\bar{V}_\pi - V_\pi\|_{2,\mu} \leq \frac{1}{\sqrt{1-\gamma^2}} \|\Pi V_\pi - V_\pi\|_{2,\mu}$.

TD Learning with Function Approximation

Tabular TD to Function Approximation

- Tabular TD learning

$$V_{t+1}(s_t) = V_t(s_t) + \alpha_t(r_t + \gamma V_t(s_{t+1}) - V_t(s_t)), \text{ and, } V_{t+1}(s) = V_t(s) \text{ for all } s \neq s_t$$

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- **Tabular TD learning**

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- Define the TD error $\delta_t(w) = r_t + \gamma V(w_t)(s_{t+1}) - V(w)(s_t)$

- Update the parameter in order to minimize the squared TD error;

$$w_{t+1} = w_t - \alpha_t \nabla_w \delta_t^2(w)|_{w=w_t}, \text{ where,}$$
$$\nabla_w \delta_t^2(w)|_{w=w_t} = -\delta_t(w_t) \nabla_w V(w)(s_t)|_{w=w_t}$$

TD Learning with Linear Function Approximation

- **TD learning with function approximation:** Define the TD error at time step t as $\delta_t(w) = r_t + \gamma V(w_t)(s_{t+1}) - V(w)(s_t)$. Update the parameter as

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- **TD learning with **linear** function approximation:** Approximate the value function as $V(w)(s) = w^\top \phi(s)$, where $\phi(s) = (\phi_1(s), \phi_2(s), \dots, \phi_d(s))^\top$ is the feature vector. Then, update the parameter as

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Theorem (Convergence of TD learning with linear function approximation)

Let w^* be such that $V(w^*) = \Pi T_\pi V(w^*)$. Then $w_t \rightarrow w^*$ almost surely, where w_t is given by the TD learning update equation above.

Matrix Notation

- We can write $V(w)(s) = (\Phi w)(s)$, where

$$\Phi = \begin{bmatrix} \phi_1(s_1) & \phi_2(s_1) & \cdots & \phi_d(s_1) \\ \phi_1(s_2) & \phi_2(s_2) & \cdots & \phi_d(s_2) \\ \vdots & & \ddots & \vdots \\ \phi_1(s_{|S|}) & \phi_2(s_{|S|}) & \cdots & \phi_d(s_{|S|}) \end{bmatrix} = \begin{bmatrix} | & | & \cdots & | \\ \phi_1 & \phi_2 & \cdots & \phi_d \\ | & | & \cdots & | \end{bmatrix}$$

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$$w_{t+1} = w_t + \alpha_t (r_t + \gamma w_t^\top \phi(s_{t+1}) - w_t^\top \phi(s_t)) \phi(s_t)$$

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ODE Approximation

- TD learning update equation

$$w_{t+1} = w_t + \alpha_t (r_t + \gamma (\Phi w_t)(s_{t+1}) - (\Phi w_t)(s_t)) \phi(s_t)$$

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- This can be written as

$$\dot{w} = \Phi^\top D_\pi (r_\pi + \gamma (P_\pi \Phi w) - (\Phi w)),$$

where $D_\pi = \text{diag}(\mu(s_1), \dots, \mu(s_{|\mathcal{S}|}))$

A Simple Result

- Define the inner product $\langle V_1, V_2 \rangle = V_1^\top D_\pi V_2$.

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Lemma

Let Π be an orthogonal projection. Then, $\langle \Pi V_1, V_2 \rangle = \langle V_1, \Pi V_2 \rangle$

A Simple Result

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Lemma

Let Π be an orthogonal projection. Then, $\langle \Pi V_1, V_2 \rangle = \langle V_1, \Pi V_2 \rangle$

Proof: For any V_1, V_2 , due to orthogonality, $\langle \Pi V_1, (V_2 - \Pi V_2) \rangle = \langle (V_1 - \Pi V_1), V_2 \rangle = 0$. We get the desired result from this. $\square$

Convergence of TD Learning with LFA

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Let $U(t) = \|w^* - w(t)\|_2^2$. We will show that $\dot{U} < 0$ for $w(t) \neq w^*$.

$$\dot{U} = -2(w^* - w)^\top \dot{w}$$

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$$\begin{aligned}\dot{U} &= -2(w^* - w)^\top \dot{w} \\ &= -2(w^* - w)^\top \Phi^\top D_\pi (r_\pi + \gamma (P_\pi \Phi w) - (\Phi w))\end{aligned}$$

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(we get (i) by the previous projection lemma)

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Proof: Let w^* be such that $\Pi T_\pi V(w^*) = V(w^*)$.

Let $U(t) = \|w^* - w(t)\|_{2,\mu}^2$. We will show that $\dot{U} < 0$ for $w(t) \neq w^*$.

$$\begin{aligned}\dot{U} &= -2(w^* - w)^\top \dot{w} \\ &= -2(w^* - w)^\top \Phi^\top D_\pi (r_\pi + \gamma (P_\pi \Phi w) - (\Phi w)) \\ &= -2(\Phi w^* - \Phi w)^\top D_\pi (T_\pi(\Phi w) - (\Phi w)) \\ &= -2(\Phi w^* - \Phi w)^\top D_\pi (T_\pi(\Phi w) - (\Phi w^*) + (\Phi w^*) - (\Phi w)) \\ &= -2 \left[(\Phi w^* - \Phi w)^\top D_\pi (\Phi w^* - \Phi w) + (\Phi w^* - \Phi w)^\top D_\pi (T_\pi \Phi w - \Phi w^*) \right] \\ &= -2 \left[\|\Phi w^* - \Phi w\|_{2,\mu}^2 + (\Phi w^* - \Phi w)^\top D_\pi (T_\pi \Phi w - \Phi w^*) \right] \\ &\stackrel{(i)}{=} -2 \left[\|\Phi w^* - \Phi w\|_{2,\mu}^2 + (\Phi w^* - \Phi w)^\top D_\pi (\Pi T_\pi \Phi w - \Phi w^*) \right], \\ &\quad \text{(we get (i) by the previous projection lemma)} \\ &\stackrel{(ii)}{\leq} -2 \left[\|\Phi w^* - \Phi w\|_{2,\mu}^2 - \|(\Phi w^* - \Phi w)\|_{2,\mu} \|(\Pi T_\pi \Phi w - \Phi w^*)\|_{2,\mu} \right] \\ &\quad \text{(we get (ii) by Cauchy-Schwarz inequality)} \\ &\leq -2 \left[\|\Phi w^* - \Phi w\|_{2,\mu}^2 - \gamma \|(\Phi w^* - \Phi w)\|_{2,\mu}^2 \right] \\ &= -2(1 - \gamma) \|\Phi w^* - \Phi w\|_{2,\mu}^2\end{aligned}$$

Convergence of TD Learning

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$$\dot{U} \leq -2(1 - \gamma) \|\Phi w^* - \Phi w\|_{2,\mu}^2$$

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Now, using the results from stochastic approximation theory, we can argue that TD learning with linear function approximation will converge to the optimal parameter w^*



Convergence of RL Algorithms for Policy Evaluation

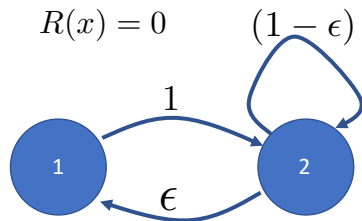
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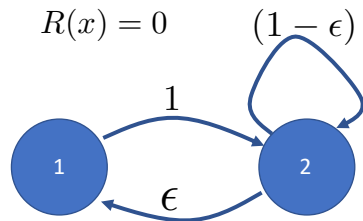
Example of Non-Convergence for Off-Policy Evaluation

- Consider the Markov chain induced by a policy π on an MDP. Clearly, $V_\pi(s) = 0$
- Use linear approximation with $\phi = \begin{bmatrix} 1 & 2 \end{bmatrix}^\top$
- We get $V(w)(s) = ws$. Optimal value of w is 0



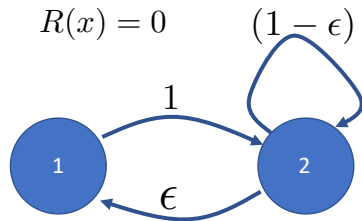
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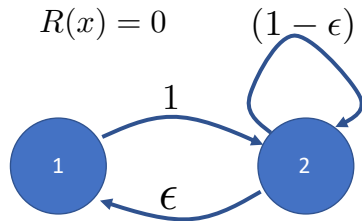
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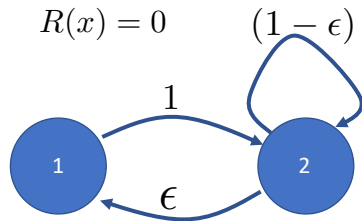
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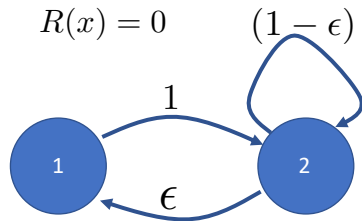
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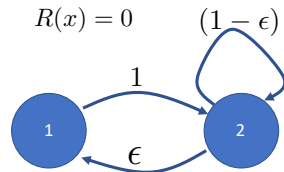
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- The distribution of the state-action pairs in the on-policy data and off-policy data will be different
- How do we perform TD learning using off-policy data obtained as above?



Example of Non-Convergence for Off-Policy Evaluation

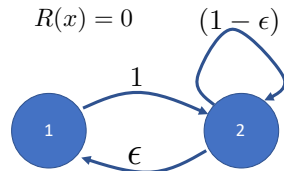
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Example of Non-Convergence for Off-Policy Evaluation

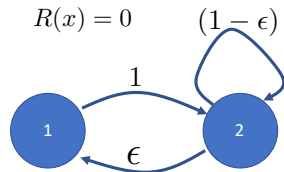
- Off-policy data: Sample state $s \in \{1, 2\}$ uniformly at random. Then get the next state according to the action taken by the policy π
- Recall TD learning with linear function approximation:

$$w_{t+1} = w_t + \alpha_t w_t (\gamma \phi(s_{t+1}) - \phi(s_t)) \phi(s_t)$$



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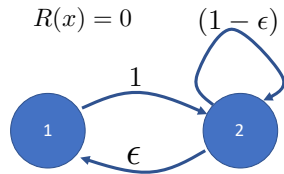


$$w_{t+1} = w_t + \alpha_t w_t (\gamma \phi(s_{t+1}) - \phi(s_t)) \phi(s_t)$$

$$\begin{aligned}\mathbb{E}[w_{t+1}] &= \mathbb{E}[w_t] + \alpha_t \mathbb{E}[w_t] \left(\frac{1}{2} (\gamma \phi(2) - \phi(1)) \phi(1) + \frac{1}{2} (\gamma (\epsilon \phi(1) + (1 - \epsilon) \phi(2)) - \phi(2)) \phi(2) \right) \\ &= \mathbb{E}[w_t] + \alpha_t \mathbb{E}[w_t] \left(\frac{1}{2} (2\gamma - 1) + \frac{1}{2} (\gamma(4 - 2\epsilon) - 4) \right) \\ &= \mathbb{E}[w_t] + \alpha_t \mathbb{E}[w_t] \frac{1}{2} (\gamma(6 - 2\epsilon) - 5)\end{aligned}$$

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- This will diverge if $\gamma > 5/(6 - 2\epsilon)$

Q-Learning with Function Approximation

Tabular Q-Learning to Function Approximation

- Tabular Q-learning

$$Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) + \alpha_t (r_t + \gamma \max_b Q_t(s_{t+1}, b) - Q_t(s_t, a_t))$$

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- Update the parameter in order to minimize the squared TD error;

$$w_{t+1} = w_t - \alpha_t \nabla_w \delta_t^2(w)|_{w=w_t}, \text{ where,}$$
$$\nabla_w \delta_t^2(w)|_{w=w_t} = -\delta_t(w_t) \nabla_w Q(w)(s_t, a_t)|_{w=w_t}$$

Q-Learning with Function Approximation

- **Q-learning with function approximation:** Define

$\delta_t(w) = r_t + \gamma \max_b Q(w_t)(s_{t+1}, b) - Q(w)(s_t, a_t)$. Update the parameter as

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- **Q-learning with linear function approximation:** Approximate the value function as $Q(w)(s) = w^\top \phi(s, a)$, where $\phi(s) = (\phi_1(s, a), \phi_2(s, a), \dots, \phi_d(s, a))^\top$ is the feature vector. Then, update the parameter as

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Convergence of Q-Learning Algorithm with Function Approximation

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"Deadly Triad" [SB, Chapter 11]

- RL algorithms shows instability and divergence whenever we combine all of the following three elements:
- **Function approximation**: a scalable way of generalizing from a state space much larger than the memory and computational resources
- **Bootstrapping**: Update targets that include existing estimates rather than relying exclusively on actual rewards and complete returns
- **Off-policy training**: training on a distribution of transitions other than that produced by the target policy